

Note 1.1 Systems of Linear Equations

Linear Equations

- **Definition:** A linear equation in n unknowns is an equation of the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where $a_1, a_2, \dots, a_n$ and b are real numbers and $x_1, x_2, \dots, x_n$ are variables.

- **Linear System:** A linear system of m equations in n unknowns is a system of the form

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\&\vdots \\a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m\end{aligned}$$

where a_{ij} 's and b_i 's are all real numbers. And the form is an $m \times n$ linear systems.

- **Consistent and inconsistent:** If a linear system has at least one solution, we say it's consistent. If not, it's inconsistent.
- **Solution Set:** The set of all solutions of a linear system.

Equivalent Systems

- **Definition:** Two systems of equations involving the same variables are said to be equivalent if they have the same solution set.
- **Three operations to obtain an equivalent system:**
 - Interchange the order of any two equations.
 - Multiple the same nonzero real number of both sides of an equation.
 - Add(or subtract) a multiple of one equation to(or from) another.

$n \times n$ Systems

- **Strict triangular form:** A system is said to be strict triangular form if in the k th equation, the coefficients of the $k - 1$ variables are all zero and the coefficient of x_k is nonzero($k = 1, \dots, n$). For example,

$$\begin{aligned}3x_1 + 2x_2 + x_3 &= 1 \\x_2 - x_3 &= 2 \\2x_3 &= 4\end{aligned}$$

In general, we can use [the upper operations](#) to try to obtain an equivalent system that is strict triangular form.

- **Matrix:** A rectangular array of numbers. A matrix having m rows and n columns is said to be $m \times n$. A matrix is said to be square if the number of rows equals to columns'.
- **Coefficient Matrix and Augmented Matrix:** A matrix that associates with the coefficients of the linear system is said to be coefficient matrix. A matrix that attach an additional matrix on the right hand is said to be augmented matrix.(denoted by $(A|B)$) For example,

$$(A|B) = \left(\begin{array}{ccc|ccc} a_{11} & \cdots & a_{1n} & b_{11} & \cdots & b_{1r} \\ \vdots & & & \vdots & & \\ a_{m1} & \cdots & a_{mn} & b_{m1} & \cdots & b_{mr} \end{array} \right)$$

and there, we use the augmented matrix like

$$\left(\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & & & \vdots \\ a_{m1} & \cdots & a_{mn} & b_m \end{array} \right)$$

- **Elementary Row Operations to the Augmented Matrix:**
 - Interchange two rows.
 - Multiply a row by a nonzero real number.
 - Replace a row by its sum with a multiple of another row.
- **Elimination:**
 - **Pivot and Pivotal Row:** The first-nonzero entry that are needed to be eliminate is called pivot. The row containing pivot is called pivotal row.
 - **Process:**
 - Choose the pivot and the pivot row.(Interchange two rows if the first entry of first row is zero)
 - Use the [elementary row operations](#) to eliminate the column containing the pivot excluding the pivot itself.
 - Change the pivot row to the next row and repeat upper process until the matrix become the strict triangular form.
 - Back substitution.
 - Remark: The determinant of a triangular matrix A equals to the product of the pivots. To a general square matrix, we can use elimination to get a triangular matrix. But the sign of the determinant will change for every row swaps.

Geometric Representation of matrix

- **2×2 System:** To a linear system

$$\begin{aligned} 2x_1 - x_2 &= \\ -x_1 + 2x_2 &= 3 \end{aligned}$$

It can be written as

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$$

and denoted by $A =$.

- **Row Picture:** It can be represented by two lines which intersect at a single point $(1, 2)$
- **Column Picture:** It can be represented as the [linear combination](#) of the column vectors of $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$.
- $n \times n$ **System:** To a linear system

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

with m equations and n unknowns.

- **Row Picture:** Every equations represents a $n - 1$ hyperplane in the n -dimensional space. If the system has a solution, it will be the common intersection point.
- **Column Picture:** It can be represented as the linear combination of the column vectors of A . The system has a solution if and only if b can be represented as the linear combination of the column vectors of A .

Section 1.1 Exercises

- [question 9th](#)
 - The equation system is equivalent to

$$\begin{aligned} -m_1x_1 + x_2 &= b_1 \\ (m_1 - m_2)x_1 &= (b_2 - b_1) \end{aligned}$$

if $m_1 \neq m_2$, then $x_1 = \frac{b_2 - b_1}{m_1 - m_2}$, and x_2 has a unique corresponding value. Then the system will have a unique solution.

- If $m_1 = m_2$, then $-m_1x_1 + x_2 = -m_2x_1 + x_2$. So if we subtract the equation (2) from the equation (1),

$$= b_1 - b_2$$

so the system will be consistent if and only if $b_1 = b_2$.

- For part (a), if $m_1 \neq m_2$, the two equations are two lines with different slope and the intersection will be unique, which is the unique solution of the system.

For part (b), if $m_1 = m_2$, the two lines will be parallel or the same line. And they have intersections only if they are the same line, which is $b_1 = b_2$.